NP = set of all decision problems whose "yes" answers are verifiable in polynomial time

Co-NP = set of all decision problems whose "no" answers are ver. in poly. time

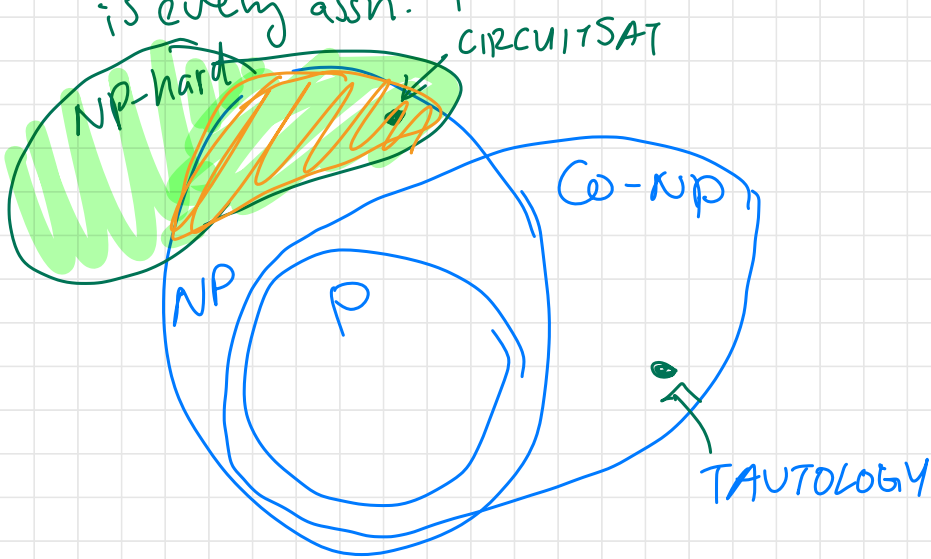
P = set of all decision solvable in polynomial time

$NP \overset{?}{=} \text{Co-}NP$

$P \overset{?}{=} NP$

CIRCUITSAT:
is there a satisfying assn?

TAUTOLOGY:
is every assn. T



NP-hard = set of all computational problems whose poly-time solvability would imply $P = NP$

Cook-Levin Theorem:

→ If CIRCUITSAT $\in P$, then $P = NP$

NP-complete: NP-hard and in NP

4color $\in P$

$\Downarrow$

3color $\in P$

$\Downarrow$

3SAT $\in P$ $\Rightarrow$ SAT $\in P$ $\Rightarrow$ CIRCUIT SAT $\in P$ $\Rightarrow$ $P = NP$

reduces to

reduce to

SAT: given a Boolean formula, is it satisfiable?

eg $(a \vee b \vee \bar{c}) \wedge (a \Rightarrow b) \vee \overline{(a = b \vee c)}$

To prove SAT is NP-hard, reduce from CIRCUITSAT in poly-time

3SAT: given a Boolean formula in ~~conjunctive~~ cnf normal form w/ 3 literals per clause, is it satisfiable?

eg $\underbrace{(a \vee b \vee c) \wedge \boxed{(b \vee b \vee c)} \wedge (\bar{c} \vee \bar{b} \vee d)}_{\text{cnf}}$

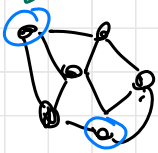
literal

clause

literal

3 clauses

To show problem is NP-hard, reduce a known NP-hard problem to it. (in poly time)



Theorem: MaxIndSET is NP-hard.

Proof: by reduction from 3-SAT.

Solve 3SAT (3CNF formula Φ w/ k clauses):

Create graph G:

for every clause, make a connected group of 3 nodes

IS can only choose at most 1 per group

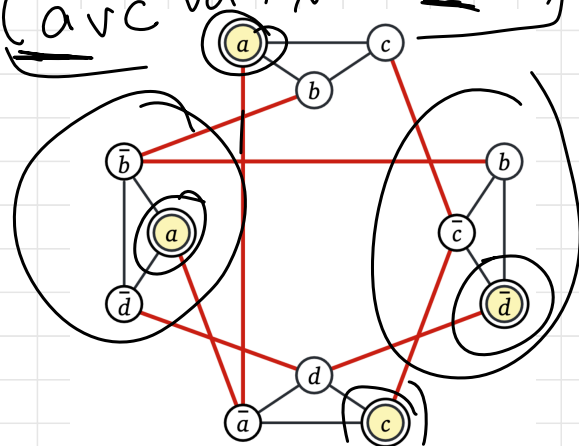
attach all x to $\bar{x}$ w/ edge

$n = \text{MaxIndSet}(G)$

can't choose x and $\bar{x}$ in IS

if $n = k$:
return T
else: return F

$(a \vee b \vee c) \wedge (b \vee \bar{c} \vee \bar{d}) \wedge (\bar{a} \vee c \vee d) \wedge (a \vee b \vee \bar{d})$



Solve 3SAT(Φ) is T <=> Φ is ~~solvable~~ satisfiable

G has a max IS of size k <=> Φ is satisfiable